TECHNISCHE UNIVERSITÄT DARMSTADT

TUTORIAL MACHINE LEARNING IN SOLID MECHANICS – TASK 2

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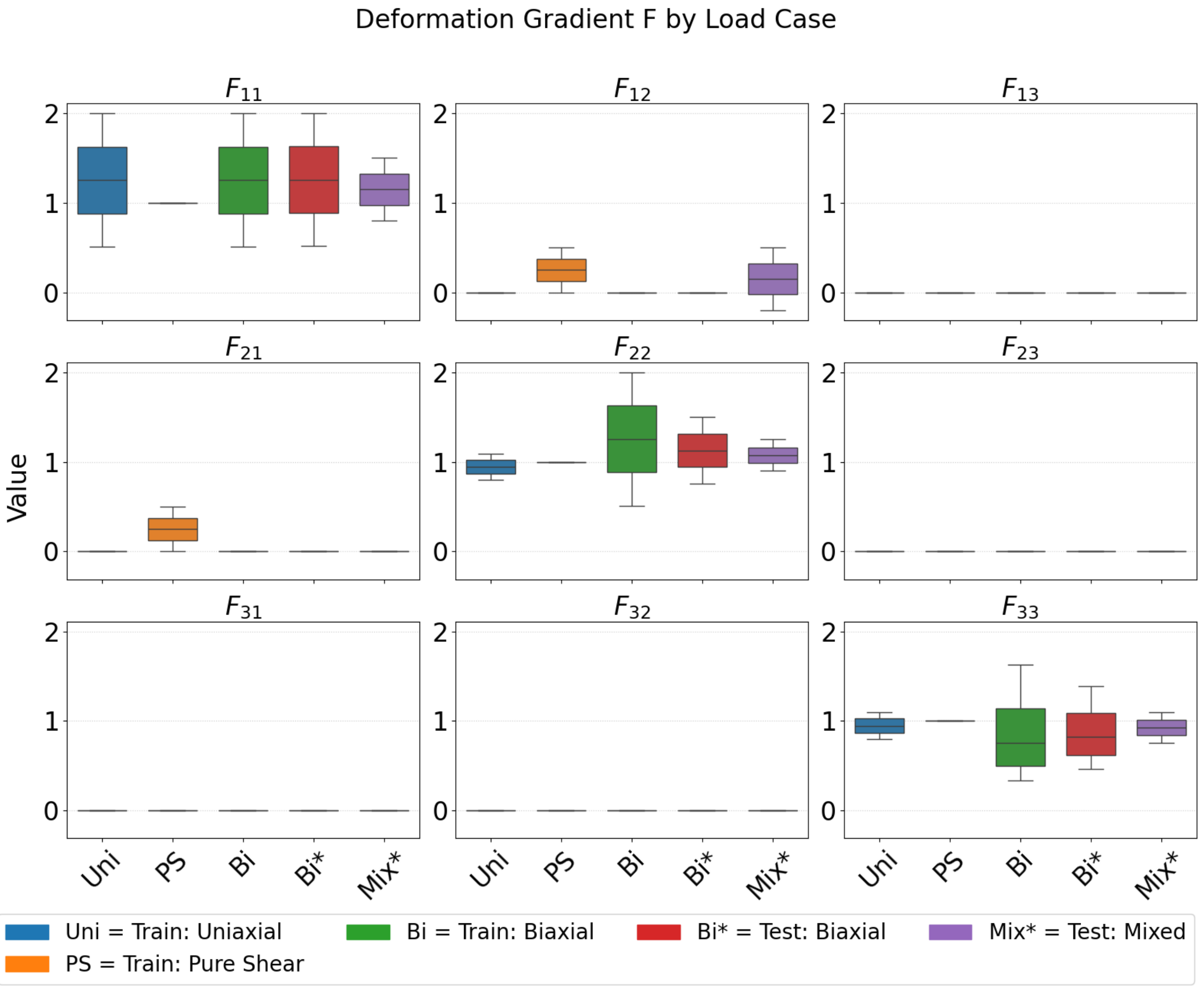
AGENDA

DATASET 1 – INTRODUCTION

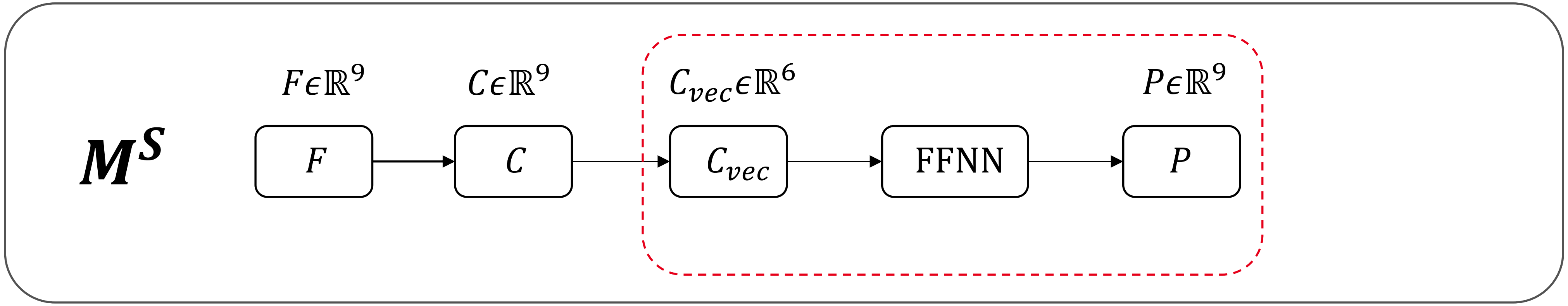
- **Transverse Isotropic Material:**

- **Training:**
  Uniaxial, Pure Shear, Biaxial.

- **Test:** Mixed Case combines stretch + shear simultaniously.
  - New combination → **Extrapolation** in feature space.



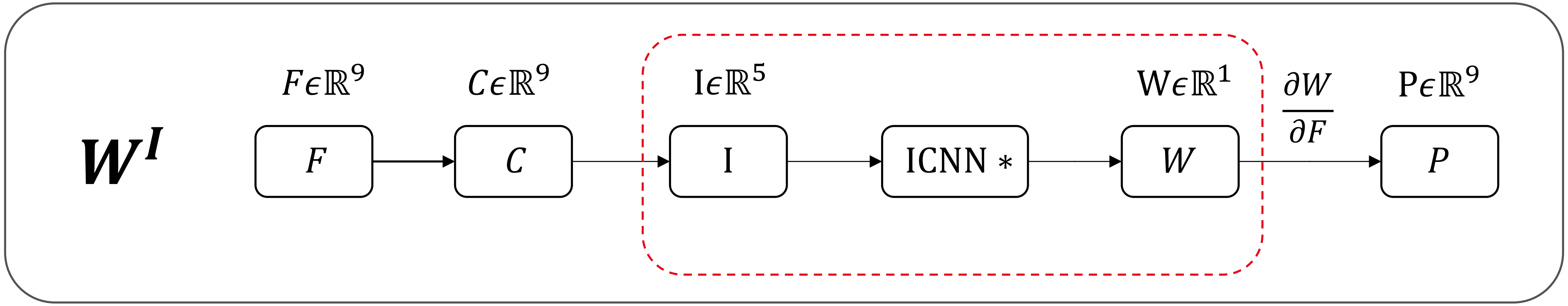
NAIVE FFNN



Constitutive Conditions:

- **Thermodynamic Consistency not guaranteed:** Stress is predicted directly.
- **Objectivity:** Fulfilled by using right Cauchy-Green Tensor C as Input.
- **Convexity not guaranteed:** Standard FFNN is used.

INVARIANT BASED PANN



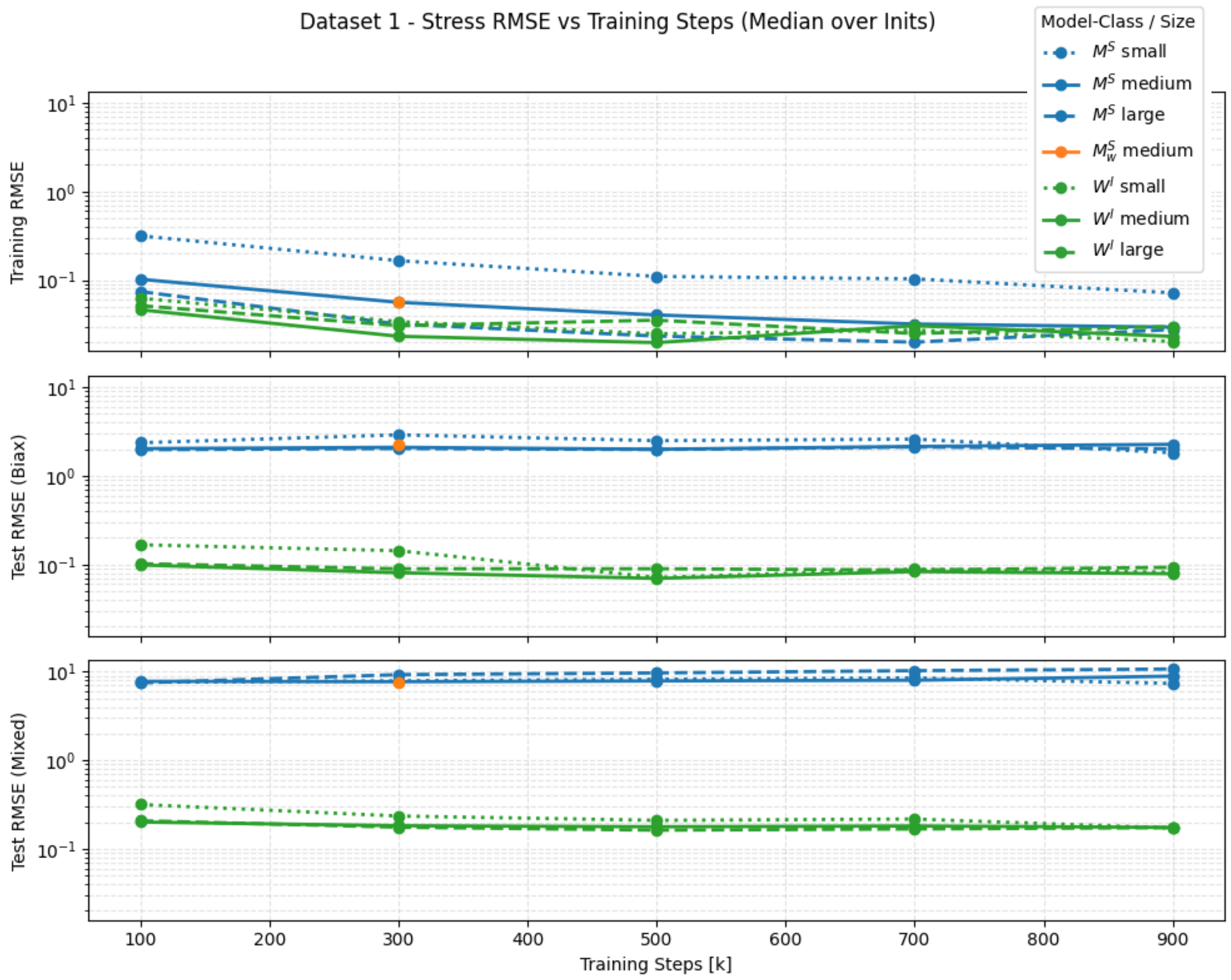
Constitutive Conditions:

- **Thermodynamic Consistency:** Via automatic differentiation ($P = \frac{\partial W}{\partial F}$).
- **Objectivity & Material Symmetry:** Enforced by invariant inputs I.
- **Polyconvexity:** ICNN* with non-negative weights in all layers.

DATASET 1 – FFNN VS. PANN

Key Observations:

- **Interpolation (Training):**
  - Both models learn to interpolate the training data (RMSE decrease).
- **Extrapolation (Test)**:
  - PANN learns to extrapolate (Test RMSE same magnitude as Train).
  - FFNN shows terrible extrapolation, especially for mixed test set.



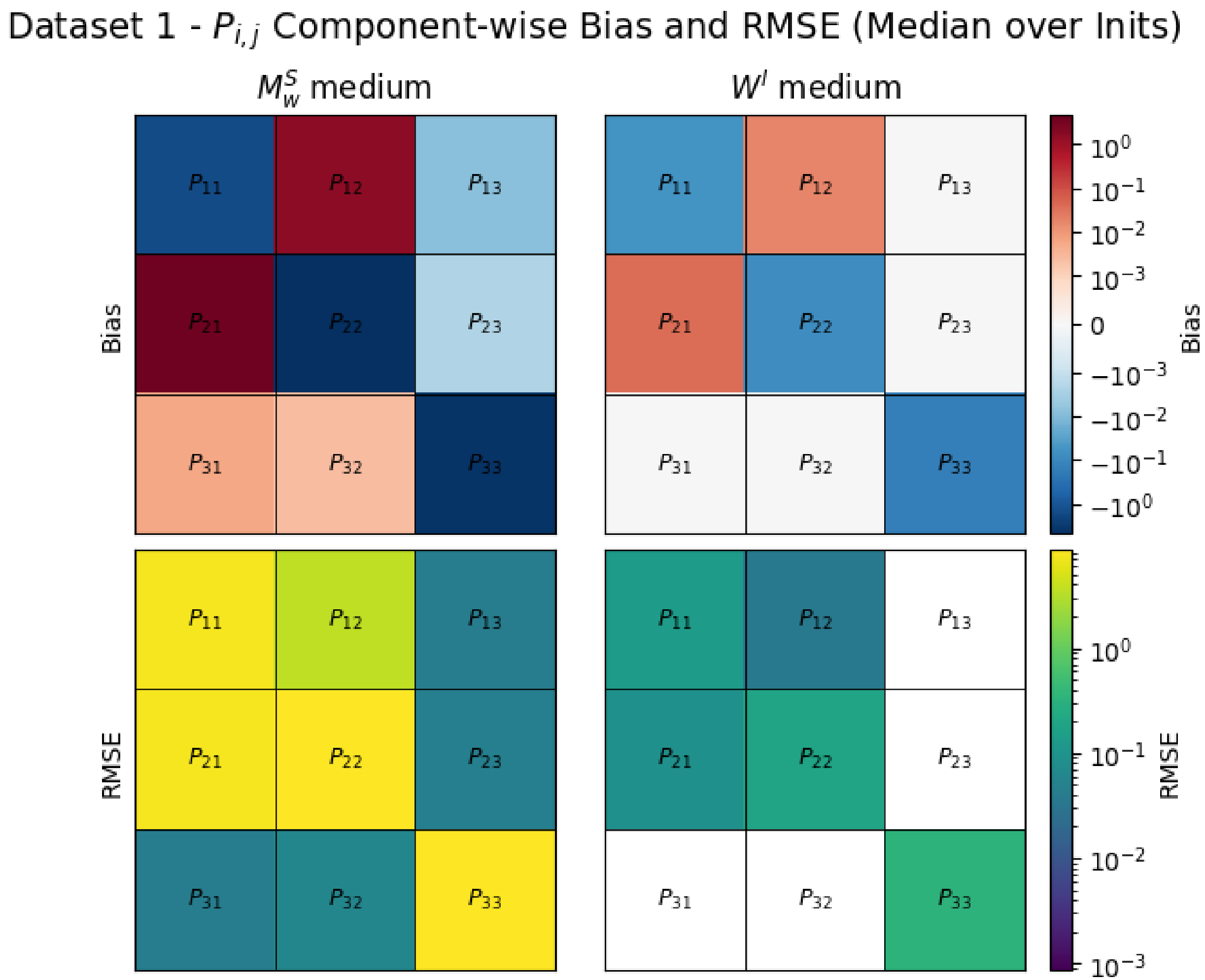
Inits = different random Initialization

$W^I = PANN(I)$, $M^S = FFNN(C_{vec})$

DATASET 1 – COMPONENT EVAL

Scope: unseen Test Data
(Biaxial + Mixed Load Cases).

- **Naive FFNN**
  - High systematic Bias and RMSE indicates incorrect physical mapping.
- **PANN**
  - Orders of magnitude lower RMSE.



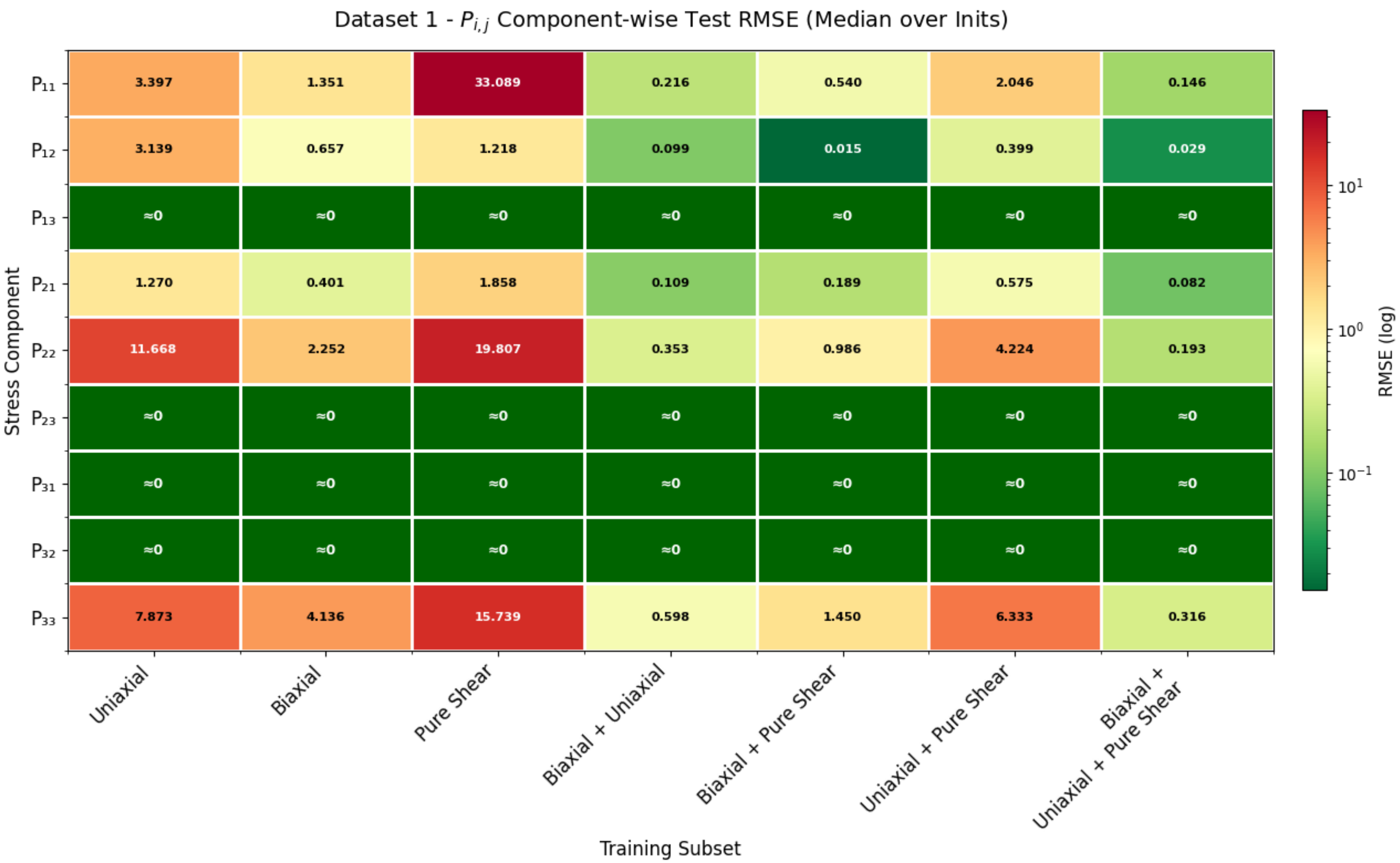
Inits = different random Initialization
$W^I = PANN(I),\ M_W^S = FFNN(C)$ uses Loss Weighted Strategy

TRAINING SET VARIATION

Scope: PANN

Key Observations:

- **Single Load Cases:** Training on single cases does not generalize well.
- **Combined Load Cases:** Combining load paths drastically improves generalization.
- **Best Performance** when trained on all datasets.



Inits = different random Initialization

$W^I = PANN(I)$

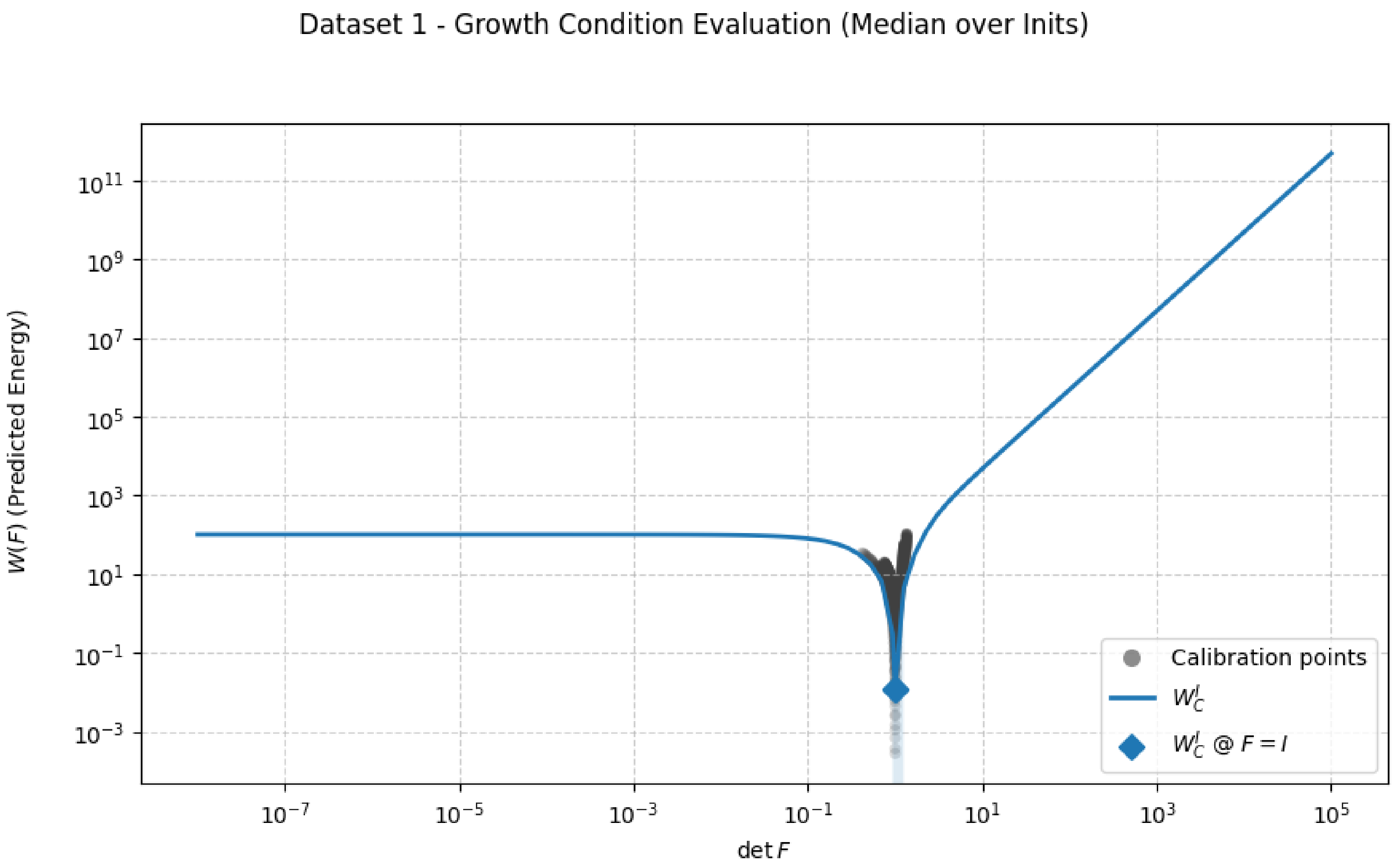
DATASET 1 – CONSTITUTIVE CONDITIONS (1)

Growth Condition:
$\psi(F) \rightarrow \infty$ as $det\ F \rightarrow 0^+, \infty$
Here: $W(F) = \psi(F)$

Key Observations:

- For $det\ F > det\ I$:
  $W(F)$ grows rapidly.
- For $det\ F < det\ I$:
  $W(F)$ approaches a constant.

➢ Growth condition not fulfilled!



Inits = different random Initialization
$W^I{}_C = PANN(I)$ that uses Sobolev Training

DATASET 1 – CONSTITUTIVE CONDITIONS (2)

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Normalization Condition:

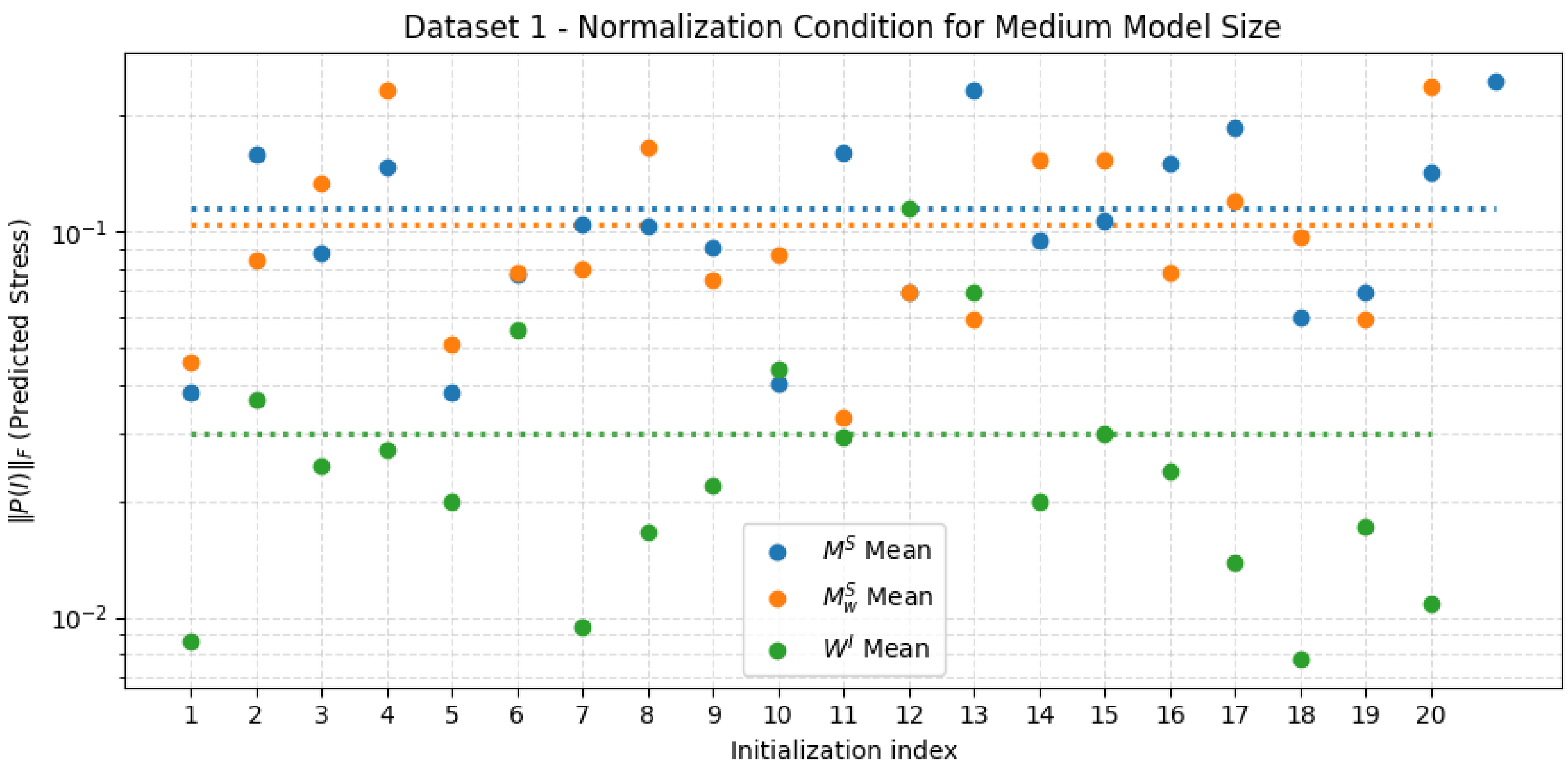
$$P(I) \ = \ 0$$

Example of $\|\cdot\|_{\mathrm{F}}$ magnitude:

$$P(I) = \|\mathrm{diag}(0{,}06)\|_{\mathrm{F}} \approx 10^{-1}$$

Key Observations:

- $P(I)$ for PANN tends to be much lower than for FFNN.
- Unstable across Initializations.



Inits = different random Initialization

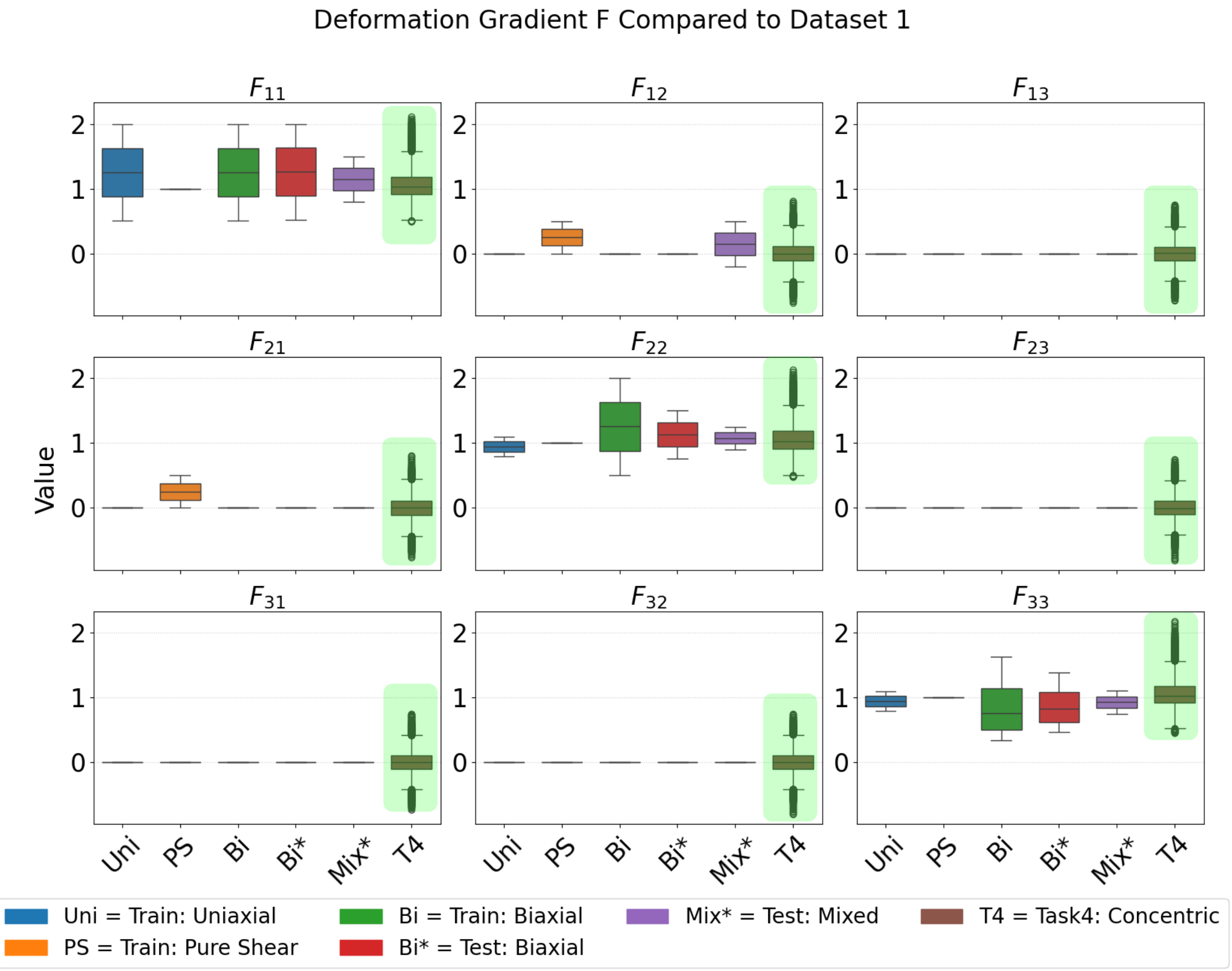
$W^I = PANN(I),\ M^S = FFNN(C),\ M^S_W = FFNN(C)$ uses Loss Weighted Strategy

DATASET 1 – KEY TAKEAWAYS

- In all tests the PANN exhibited significantly **lower RMSE** than the naïve FFNN.
- PANN benefits from **diverse training set**.
- **Growth condition** not fulfilled by PANN.
- PANN satisfies **normalization condition** more closely than FFNN.

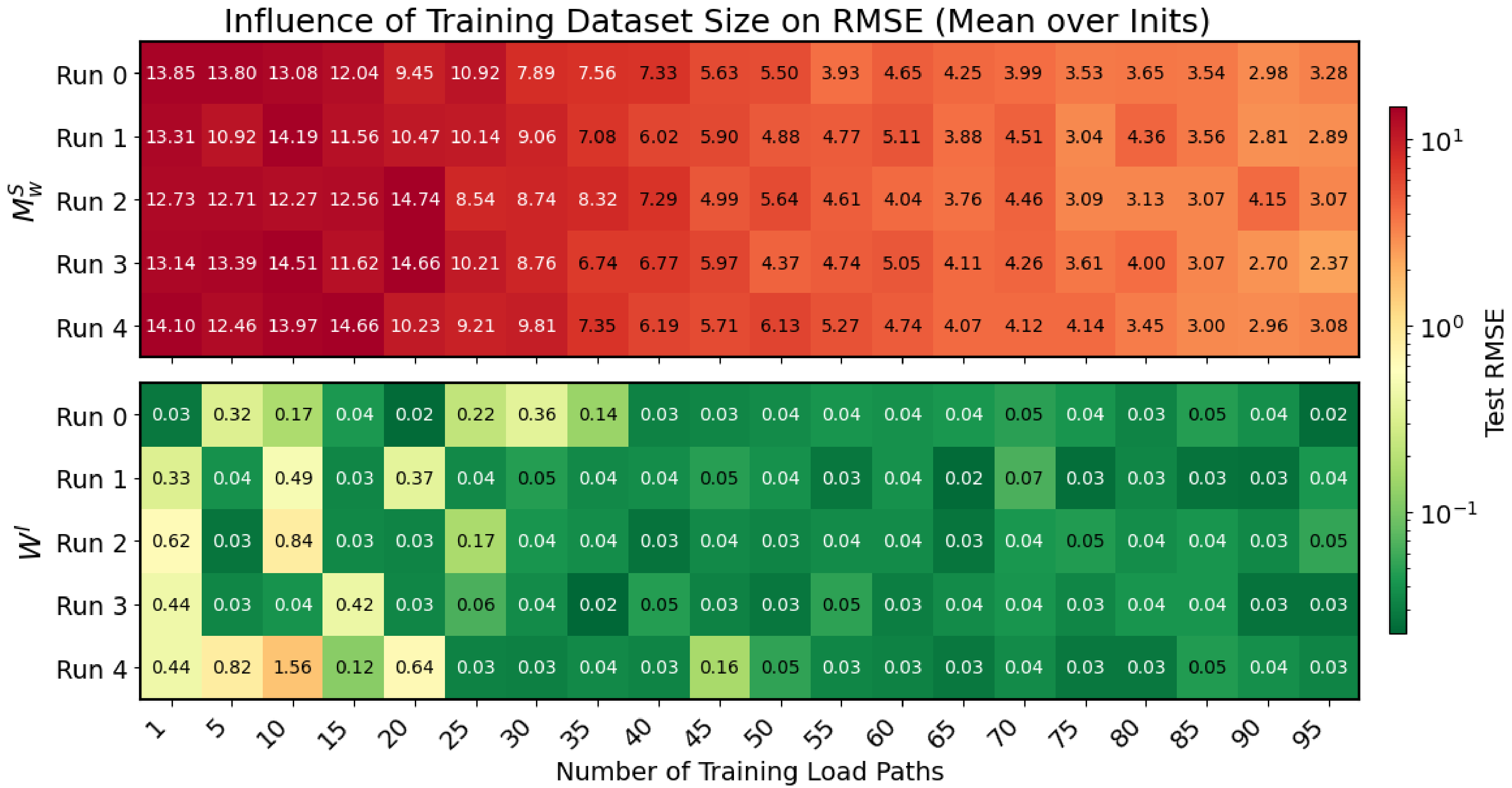
DATASET 2

- **Concentric sampling:**
- 5.000 deformation gradients (vs. 600 in Dataset 1).
- Covers all 9 components of P including off-diagonal shear.
- Enables comparison of FFNN vs. PANN with more training data.



Inits = different random Initialization
$W^I = PANN(I)$, $M^S = FFNN(C)$

EVALUATION ON DATASET 2



Inits = different random Initialization
$W^I = PANN(I)$, $M_W^S = FFNN(C)$ uses Loss Weighted Strategy

DATASET 2 – KEY TAKEAWAY

- PANN consistently provides much better predictions then naïve FFNN, especially for **small datasets**.

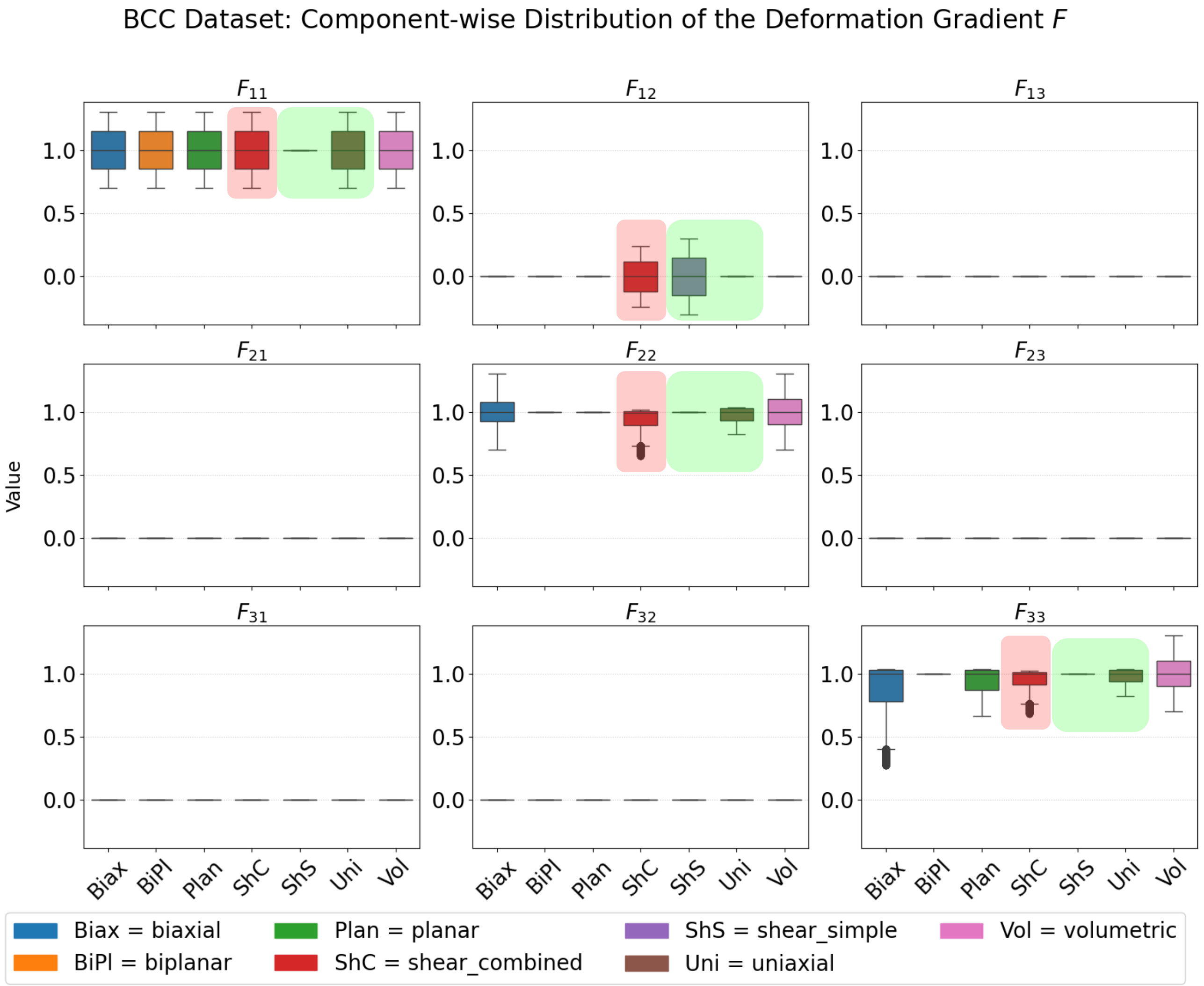
DATASET 3

- **Cubic Anisotropic Material:**

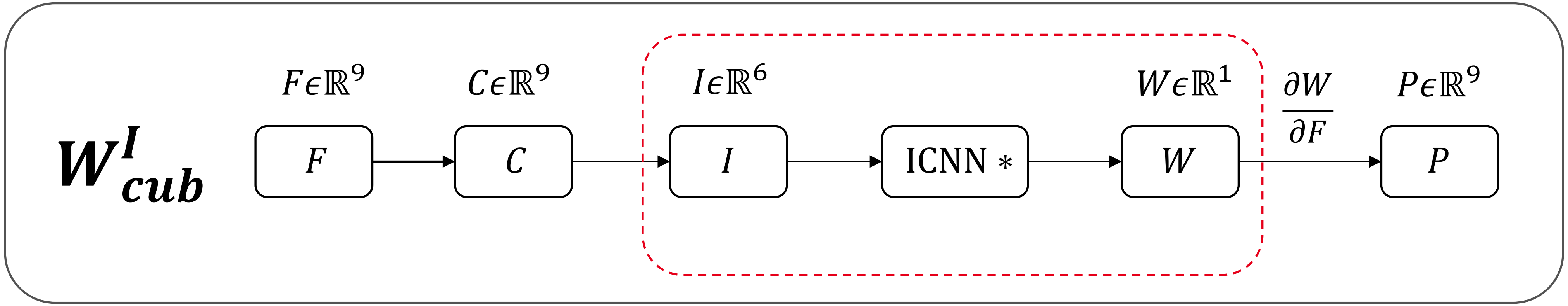
- 7 Load Cases
  - 2,060 Load Paths available.
  - 414k Total Datapoints.

- **Training:** Only basic load cases (Uniaxial & Simple Shear).

- **Test:** Mixed Shear-Tension (Shear Combined).

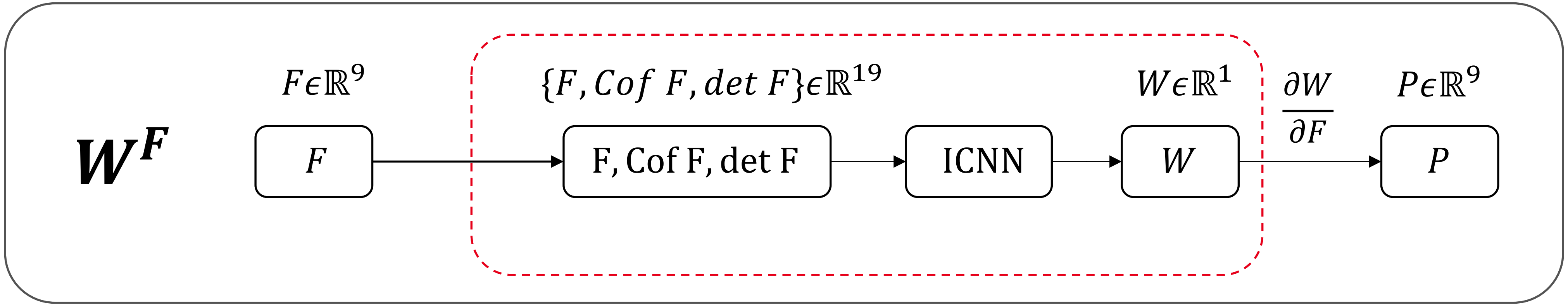


CUBIC INVARIANT BASED PANN



- **Thermodynamic Consistency:** Via automatic differentiation ($P = \frac{\partial W}{\partial F}$).
- **Objectivity & Material Symmetry:** Enforced by invariant inputs I.
- **Polyconvexity:** ICNN* with non-negative weights in all layers.

DEFORMATION GRADIENT PANN



- **Thermodynamic Consistency:** Via automatic differentiation ($P = \frac{\partial W}{\partial F}$).
- **Objectivity :** Not guaranteed. Learned via data augmentation.
- **Polyconvexity:** Guaranteed by convex function of inputs ($F, Cof\ F, \det F$).
  - **Weight Constraint relaxed:** First hidden layer can have negative weights.

DATASET 3 – MODEL EVALUATION

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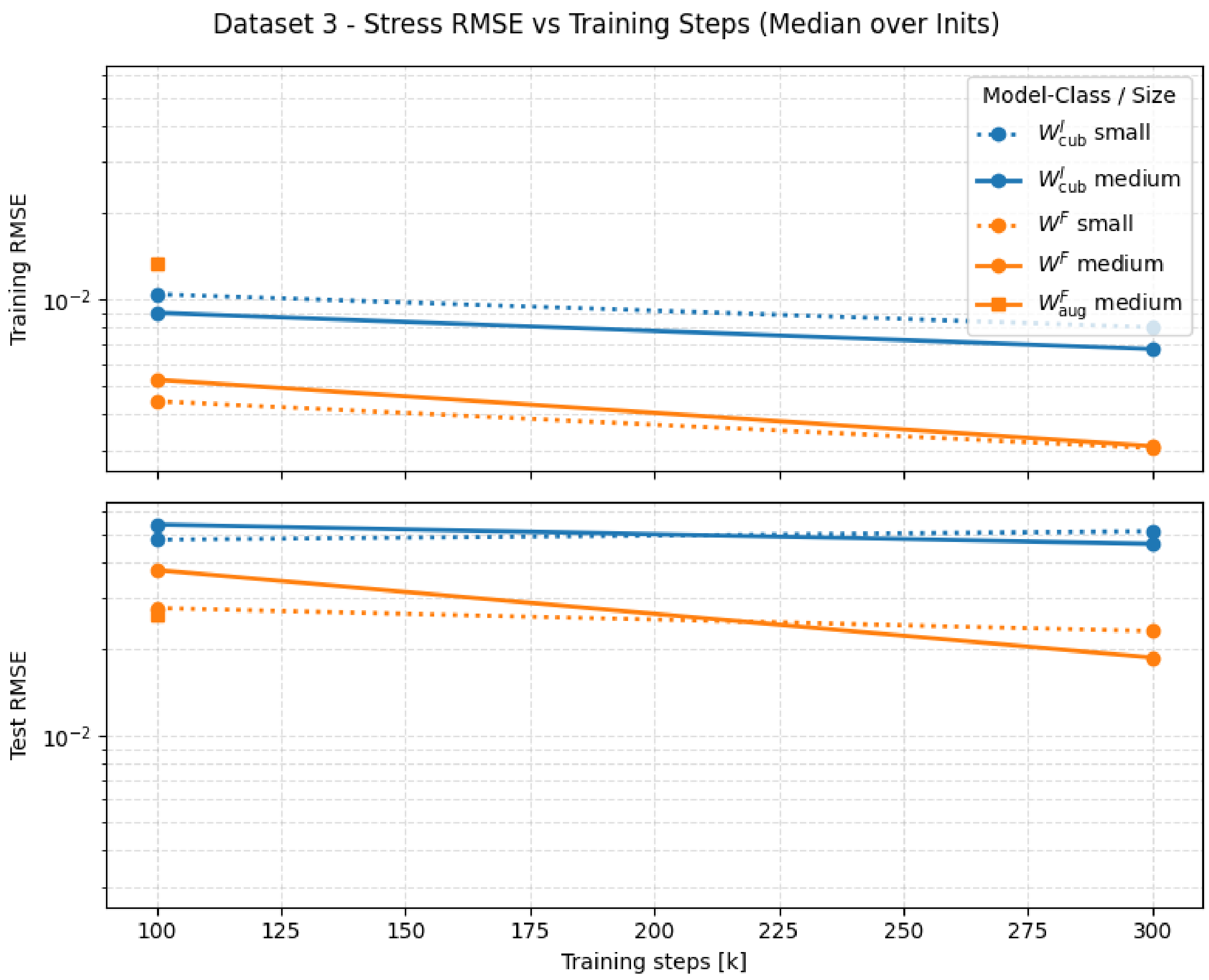
Key Observations:

- Both PANNs exhibit low RMSE on training and test sets.
- W^F extrapolates better than $W^I{}_{\mathrm{cub}}$.
- $W^F{}_{aug}$ only differs slightly in RMSE from W^{F}.

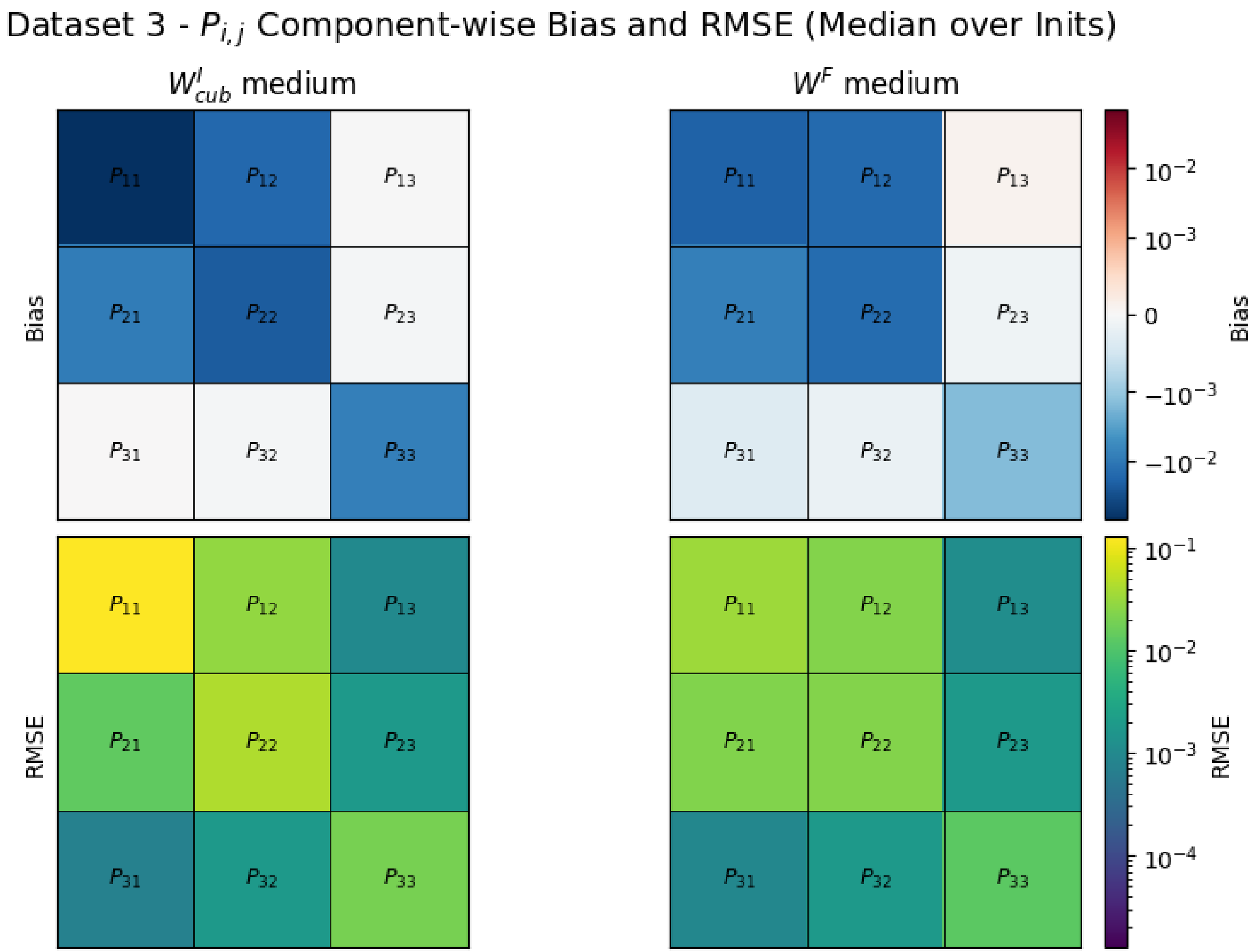


Inits = different random Initialization
$W^I{}_{cub} = PANN(I), W^F = PANN(\{F, cof\ F, det\ F\}),$
$W^F{}_{aug} = PANN(\{F, cof\ F, det\ F\})$ that uses data augmentation

DATASET 3 – COMPONENT EVAL

Scope: Test Data (Mixed Shear-Tension).

- Both PANNs show similar RMSE magnitude and distribution over $P_{i,j}$.
- Both PANNs underestimate stresses.



Inits = different random Initialization
$W^I{}_{cub} = PANN(I),\ W^F = PANN(\{F, cof\ F, det\ F\}),$

DATASET 3 – CONSTITUTIVE CONDITIONS (1)

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Growth Condition:

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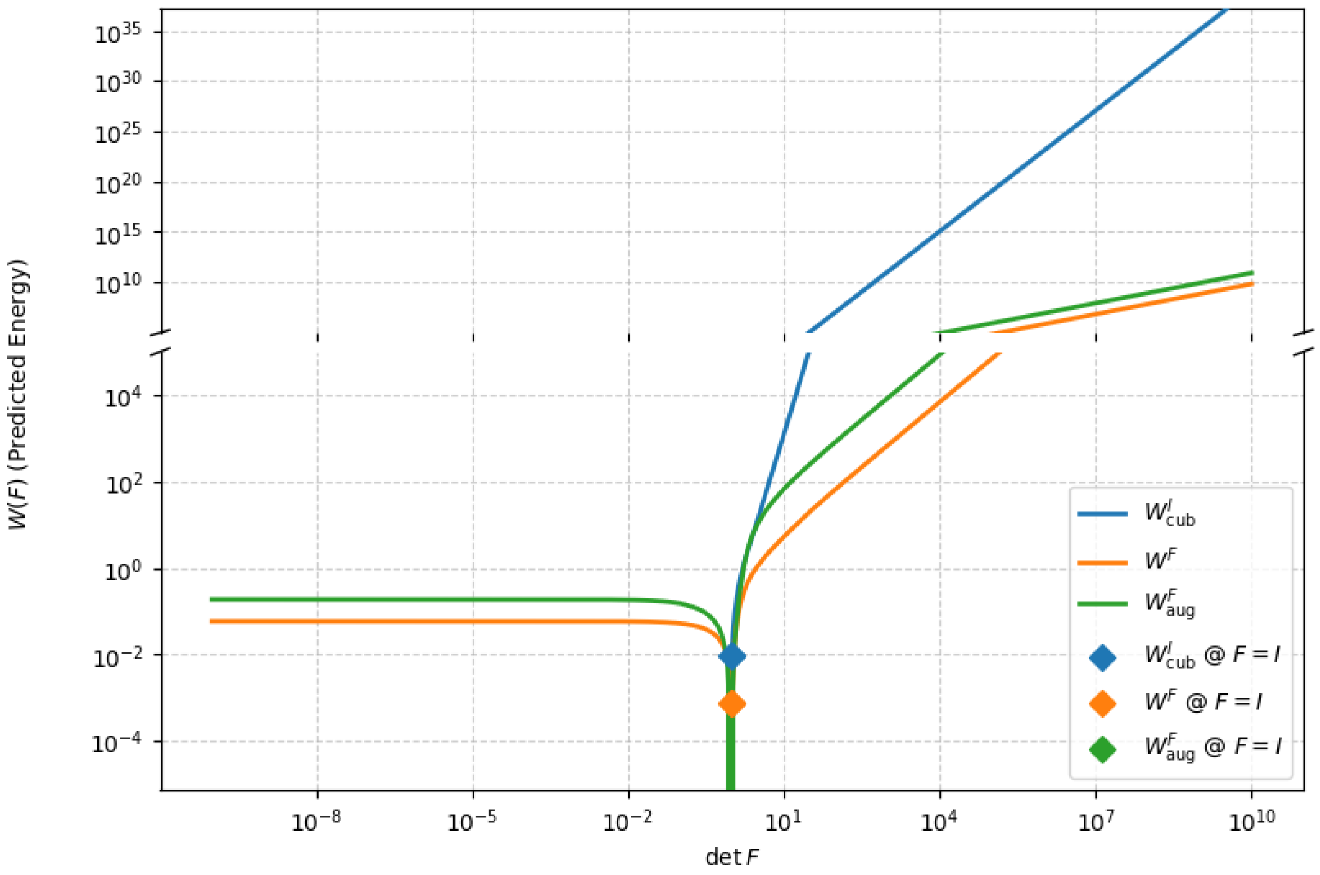
Key Observations:

- For $det\ F > det\ I$:
  $W(F)$ grows rapidly.
- For $det\ F < det\ I$:
  $W(F)$ approaches a constant.
- Growth condition not fulfilled!

Dataset 3 - Growth Condition Evaluation (Median over Inits)



Inits = different random Initialization
$W^I{}_{cub} = PANN(I), W^F = PANN(\{F, cof\ F, det\ F\}),$
$W^F{}_{aug} = PANN(\{F, cof\ F, det\ F\})$ that uses data augmentation

DATASET 3 – CONSTITUTIVE CONDITIONS (2)

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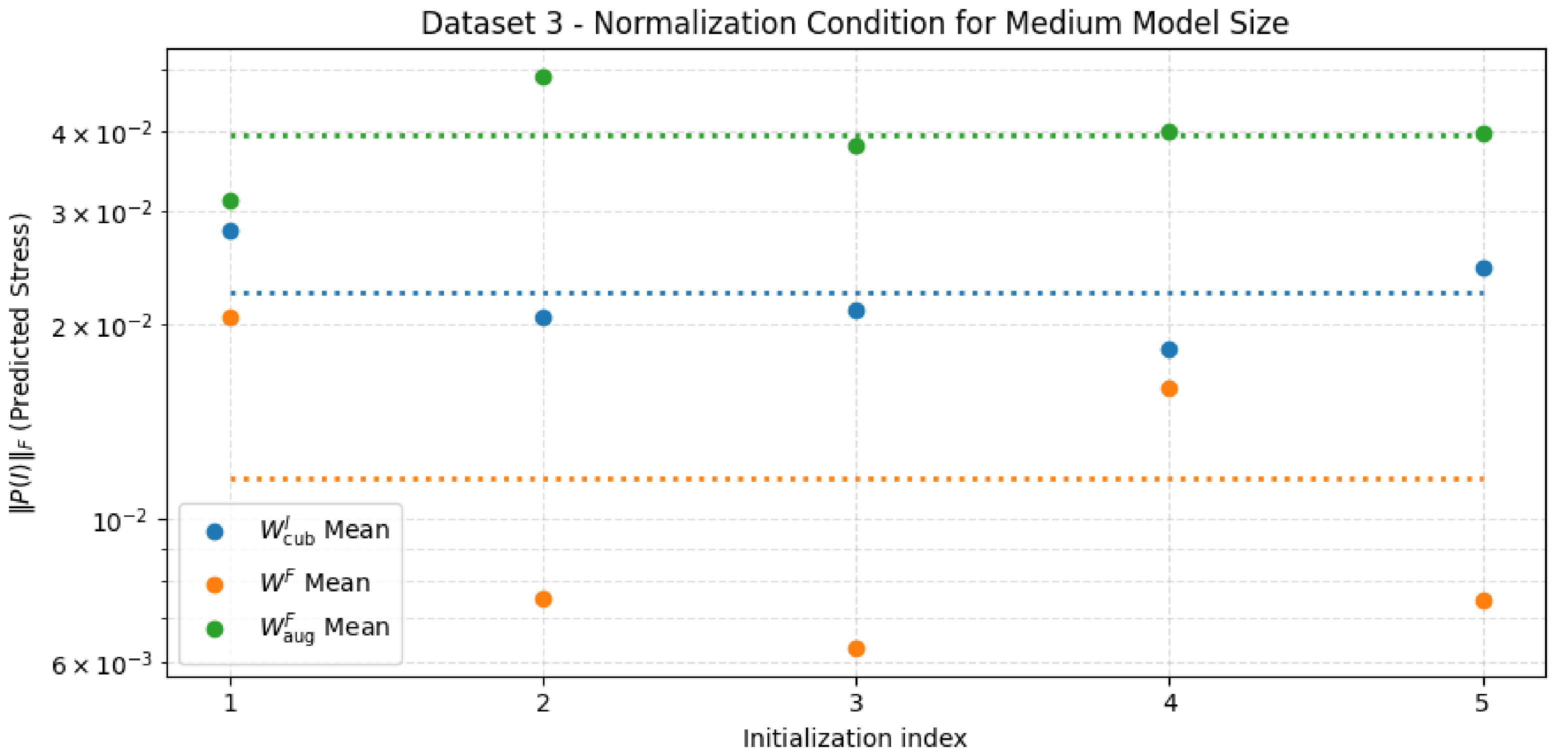
Normalization Condition:

$P(I) \;=\; 0$

Example of $\|\cdot\|_{\mathrm{F}}$ magnitude:
$P(I) = \|\mathrm{diag}(0{,}006)\|_{\mathrm{F}} \approx 10^{-2}$

Key Observations:

- $P(I)$ is very small for all PANNs.
- Normalization condition is being approximately learned.



Inits = different random Initialization
$W^I{}_{cub} = PANN(I), W^F = PANN(\{F, cof\ F, det\ F\}),$
$W^F{}_{aug} = PANN(\{F, cof\ F, det\ F\})$ that uses data augmentation

DATASET 3 – DATA AUGMENTATION

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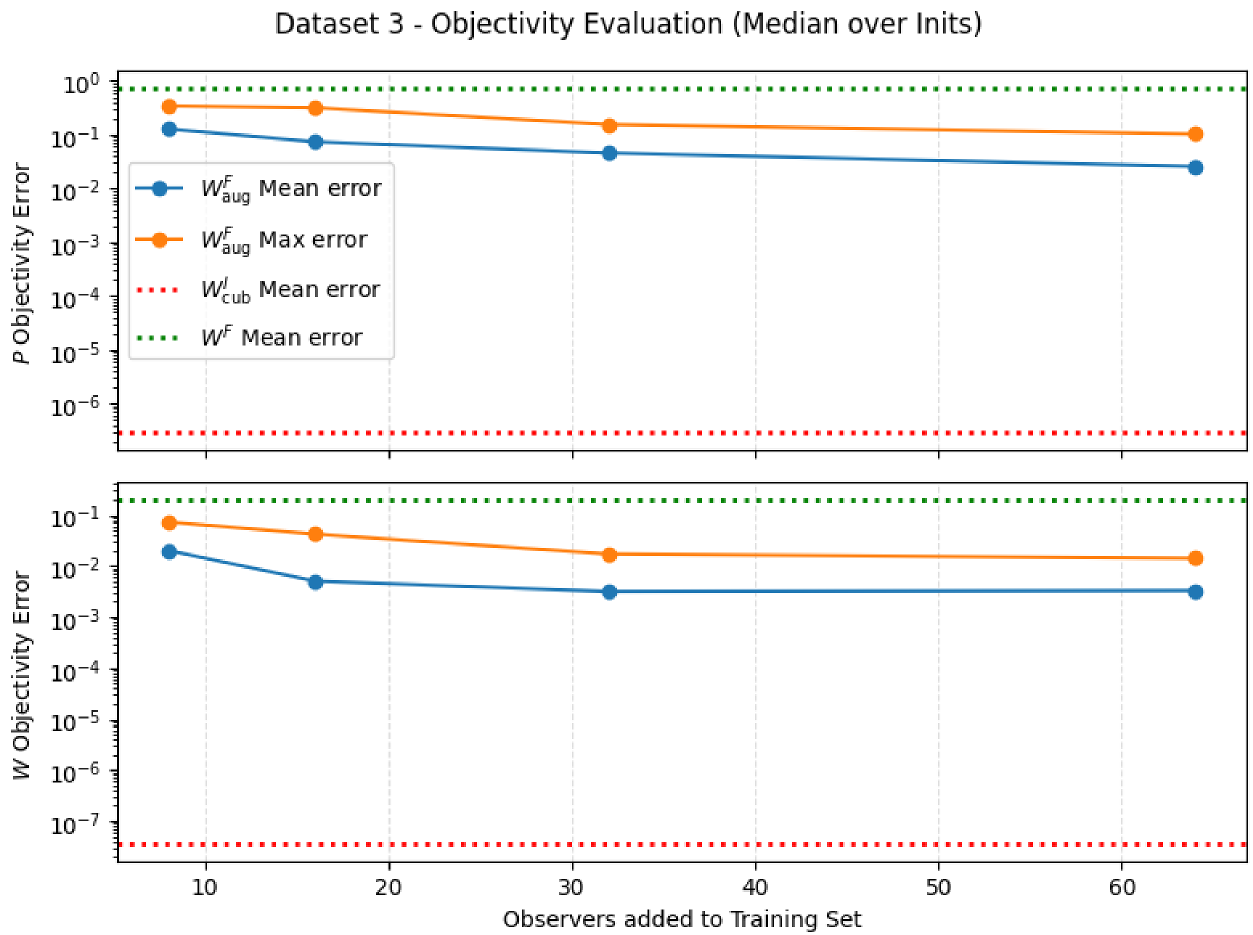
Objectivity Errors Computed as:

$$P_{err} = \|P(QF) - QP(F)\|,$$
$$W_{err} = |W(QF) - W(F)|$$

Key Observations:

- Increasing number of Q_{obj} reduces objectivity error, but not indefinitely.
- Objectivity is being approximately learned, but still much weaker than ${W^I}_{cub}$.



Inits = different random Initialization
${W^I}_{cub} = PANN(I), W^F = PANN(\{F, cof\ F, det\ F\}),$
${W^F}_{aug} = PANN(\{F, cof\ F, det\ F\})$ that uses data augmentation

DATASET 3 – KEY TAKEAWAYS

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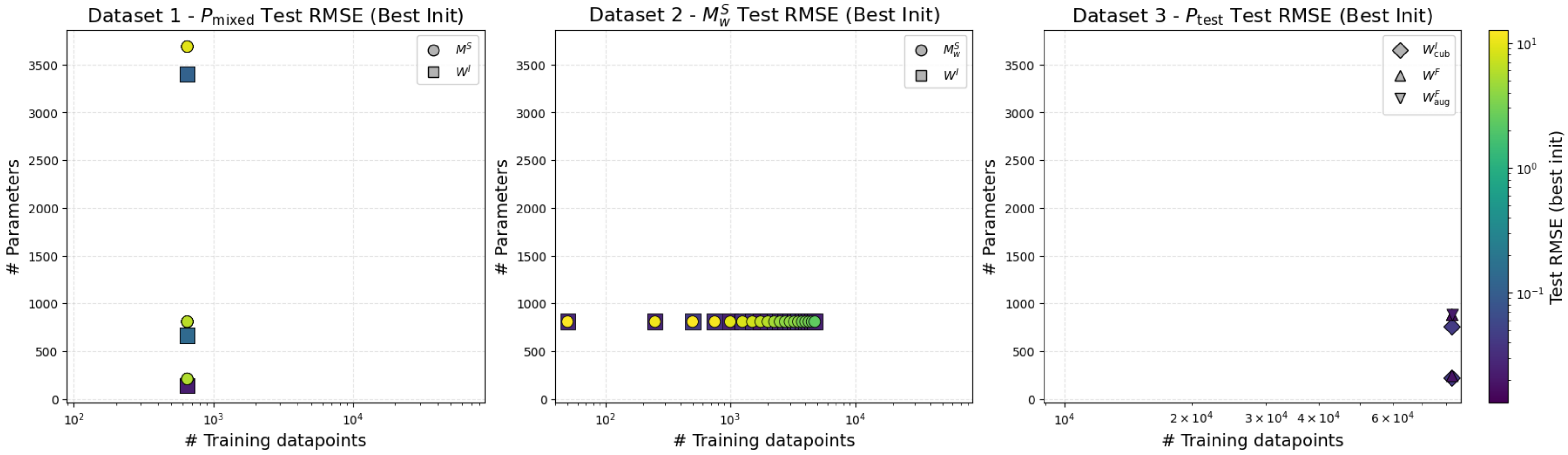
- Neither Invariant based- nor deformation gradient based PANNs learn **growth condition!**
- **Objectivity** can be learned to a degree through data augmentation.
- Learning **normalization condition** is improved with a larger training dataset!

CONCLUSION

- **PANN outperforms FFNN** consistently across all datasets, especially with limited training data.
- **Normalization** and **objectivity** can be approximately learned, **growth condition** not fulfilled.



LIMITATIONS

- **Cross-Model Comparability:**
  - Use of distinct material data limits comparability across datasets.
- **Statistical Robustness:**
  - Limited random initializations $(n_{inits} = 5)$ for Dataset 3 experiments reduce certainty of observations.
- **Experimental Scope:**
  - Training duration and augmentation intensity exploration was limited due to computational ressources.